Harish Manoharan

San Francisco Bay Area | harish_manoharan@outlook.com
linkedin.com/in/harishm17 | github.com/harishm17 | harishmanoharan.com

Software engineer at Purgo AI since June 2026 (intern from June 2025), working on agents, retrieval and LLM evaluation. M.S. Computer Science, UT Dallas, 2026 (GPA 3.92/4.00); dual degree in Data Science and Biological Sciences, IIT Madras.

EXPERIENCE

Purgo AI San Francisco Bay Area (Los Altos, CA)

AI startup building an agent that writes and validates Databricks and dbt code

Software Engineer Jun 2026 – Present

Software Engineer Intern Jun 2025 – May 2026

- Cut the LangGraph agent's missed source tables by 70% (79 to 24 on 103 tickets) by building dbt lineage expansion with sqlglot; chose the design in a follow-up four-variant ablation and rejected repo search (4.7x more unwanted tables).
- Ran a 5-run, 102-ticket comparison showing a smaller reasoning model lost no retrieval recall (0.944 vs 0.931 for the larger model), then moved the agent's retrieval and drafting stages to it; those two stages became 7.6x cheaper.
- Built Pinecone semantic search for Polaris/Iceberg catalogs (incremental indexing) and its 188-query benchmark: 0.18 s mean latency vs 4.0 s and recall@10 of 0.68 vs 0.54 for the existing search path.
- Built an LLM evaluation and validation product end to end that qualifies Azure OpenAI and Databricks-served models for GxP use: 19 tests (hallucination on missing fields, accuracy/F1, bias, determinism, latency under concurrency, traceability).
- Built the IQ/OQ (installation and operational qualification) engine in TypeScript for Databricks on Azure and AWS: 59 test definitions with per-test timeouts, auditor evidence reports and an LLM pipeline turning requirements into tests.
- Extended the self-review loop in the agent's LangGraph orchestration: AST lint rules for generated PySpark/SQL, each traced to a real Databricks runtime failure, plus recovery regeneration and best-candidate selection.

Digitus Business Solutions Bengaluru, India

Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building a text-to-SQL agent with schema discovery and retries driven by execution errors.
- Built a multimodal RAG assistant over internal slide decks and videos (layout-aware chunking, enforced citations).

PwC India Bengaluru, India

Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble; built an aspect-based sentiment pipeline over 30k+ customer reviews.

EDUCATION

The University of Texas at Dallas M.S. Computer Science May 2026

GPA 3.92/4.00 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024

INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), embeddings and vector search (Pinecone, pgvector), LLM observability (Langfuse), PyTorch

Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg

Backend and cloud: FastAPI, NestJS, Next.js, REST APIs, Azure, AWS, GCP, Docker, CI/CD (GitHub Actions)